CR23532 Cyber and Digital Forensics – Lab Record

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Screenshot spaces are marked *[Fig X.Y]*. Each figure should be captured on your own system during the experiment.

Experiment 1 – Study of Computer Forensics and Different Tools Used for Forensic Investigation

AIM:

To understand the concept of computer forensics, the steps followed in a forensic investigation, and the purpose of the main tools used by investigators.

SOFTWARE REQUIREMENTS:

- OS: Windows or Linux
- Tools: Autopsy, FTK Imager, The Sleuth Kit and similar forensic utilities

THEORY:

Computer forensics is a branch of forensic science that deals with digital evidence. It covers how evidence on computers, phones and storage media is found, acquired, safeguarded, inspected and explained. It is applied in cases of hacking, data theft, policy violations and other incidents involving computers.

Investigators study hard disks, OS records, logs, web history, deleted data and similar sources to uncover facts about an incident.

The key requirement throughout is that evidence must stay unchanged, so that the conclusions drawn from it can be trusted.

PHASES OF DIGITAL FORENSICS:

1. **Identification** – Deciding which devices and records (disks, pen drives, phones, logs) may hold evidence.
2. **Collection** – Acquiring that data with proper forensic methods.
3. **Preservation** – Keeping the evidence safe from changes; images are made and hash values recorded.
4. **Examination** – Looking through the data for files, metadata and deleted items of interest.
5. **Analysis** – Connecting the findings to build a sequence of events.
6. **Presentation** – Reporting the procedure, findings and conclusions clearly.

TOOLS USED IN COMPUTER FORENSICS:

A. EnCase Forensic – Paid forensic software for acquiring and analysing data from computers and storage devices.

[Fig 1.1: EnCase interface]

B. Forensic Toolkit (FTK) – A suite that indexes evidence for fast searching and detailed file analysis.

C. The Sleuth Kit (TSK) – Open-source command-line programs for studying file systems and disk images.

D. Autopsy – A user-friendly GUI over The Sleuth Kit for disk image analysis, file recovery and keyword searching.

[Fig 1.2: Autopsy interface]

E. FTK Imager – A tool for making forensic images of drives and previewing their contents.

[Fig 1.3: FTK Imager interface]

APPLICATIONS:

1. Investigating cybercrimes such as hacking and malware attacks.
2. Recovering lost or deleted data.
3. Responding to and analysing security incidents.
4. Supporting authorised internal inquiries into employee misuse.
5. Providing evidence for legal cases.

6. Studying network logs to trace suspicious traffic.

PROCEDURE:

1. Learn the basic ideas behind digital forensics.
2. Understand each phase of a forensic investigation.
3. Study commercial and open-source forensic tools.
4. Understand what FTK Imager, Autopsy and The Sleuth Kit are used for.
5. Study real-world uses of digital forensics.
6. Note down the observations.

OBSERVATION:

The basics of computer forensics and the role of digital evidence were understood. The listed tools were compared by purpose and features. The investigation was seen to move through identification, collection, preservation, examination, analysis and presentation.

RESULT:

Thus, computer forensics and the various tools used for forensic investigation were studied successfully.

Experiment 2 – Live Forensics Case Investigation Using Autopsy

AIM:

To investigate digital evidence with Autopsy by setting up a case, attaching a data source, running the analysis and reviewing the generated artifacts.

SOFTWARE REQUIREMENTS:

- OS: Windows
- Software: Autopsy
- Evidence: disk image or logical files
- Hardware: system with adequate disk space and memory

THEORY:

Autopsy is an open-source forensic platform with a graphical interface, powered by The Sleuth Kit. It helps examiners explore file systems, recover deleted items, run keyword searches and view user activity.

An investigation in Autopsy starts with a **case**, which holds case details and evidence. A **data source** (disk image, local disk or logical files) is then added. Autopsy runs the selected **ingest modules** on it and organises the results into views.

These results reveal files, metadata, web history, accounts and other useful artifacts.

PROCEDURE:

Step 1: Start Autopsy

1. Install Autopsy from the official website.
2. Open the program.
3. Choose **New Case** on the start screen.

[Fig 2.1: Welcome window]

Step 2: Enter case information

1. Give the case a name (for example, `AP_Forensic_Case`).
2. Select a base directory for the case files.
3. Press **Next**.

[Fig 2.2: Case Information page]

Step 3: Enter optional information

1. Type a case number.
2. Enter the examiner name as **Anu Priya P**.
3. Write a short note about the case.
4. Press **Finish**.

[Fig 2.3: Optional Information page]

Step 4: Add a data source

1. Select the add data source option.
2. Choose the type of source (Disk Image/VM File, Local Disk or Logical Files).
3. Locate the evidence on the system.
4. Select it and go to the next step.

[Fig 2.4: Data source type selection]

Step 5: Choose ingest modules

1. View the list of ingest modules.
2. Enable the ones needed for the case.
3. Press **Next**.
4. Confirm and finish adding the source.

[Fig 2.5: Ingest module configuration]

Step 6: Run the analysis

1. Finish the wizard.

2. Let Autopsy analyse the evidence.
3. Wait until it completes.
4. Look at the results in the main window.

[Fig 2.6: Data source added confirmation]

Step 7: Review the artifacts

Explore File Views, Data Artifacts and Analysis Results. Clicking a file shows its metadata such as size, MIME type, timestamps, MD5 and SHA-256. The artifacts found depend on the evidence and the modules chosen.

[Fig 2.7: File listing with metadata]

HASH VERIFICATION (SHA-256):

The integrity of the evidence was rechecked in PowerShell after analysis:

```
Get-FileHash "<evidence file path>" -Algorithm SHA256
```

[Fig 2.8: Hash output]

OBSERVATION:

A case was set up in Autopsy and a data source was attached. The ingest modules analysed the evidence and listed artifacts for review. The results shown depended on the content of the evidence. The SHA-256 hash confirmed the evidence was unaltered.

RESULT:

Thus, a forensic investigation was carried out using Autopsy by creating a case, adding a data source, analysing the evidence and examining the artifacts.

Experiment 3 – Recover Deleted Files Using Forensic Tools

AIM:

To retrieve a deleted file from a forensic image using forensic tools and verify the recovered content.

SOFTWARE REQUIREMENTS:

- OS: Windows
- FTK Imager
- Recover My Files
- Test USB drive

THEORY:

When a file is deleted, the operating system normally removes only its entry and marks the space as available. The actual contents stay on the disk until other data replaces them, so recovery is often possible.

FTK Imager creates a bit-by-bit forensic image of the device, and a recovery tool scans that image for deleted files. Examining the image instead of the original device keeps the original evidence safe.

PROCEDURE:

Step 1: Prepare a sample file

1. Open Notepad.
2. Enter sample details such as a case ID, your name **Anu Priya P**, the date and a few keywords.
3. Save the file as `deleted_secret.txt` on the test USB drive.

[Fig 3.1: Sample file contents]

Step 2: Delete the file

1. Locate `deleted_secret.txt`.
2. Right-click and select **Delete**.
3. Verify that the file is gone from its folder.

[Fig 3.2: File deletion]

Step 3: Acquire an image using FTK Imager

1. Open FTK Imager.
2. Click **File → Create Disk Image**.
3. Select Physical Drive as the source.
4. Pick the test USB drive.
5. Fill in case number, evidence number and examiner.
6. Set the destination folder and choose E01 format.
7. Start imaging and let it complete and verify.

[Fig 3.3: Imaging in progress]

[Fig 3.4: Verify results with matching hashes]

[Fig 3.5: Image summary]

Step 4: Recover and check the file

1. Scan the image with the recovery tool.
2. Restore `deleted_secret.txt` to a different folder.

3. Open it and compare it with the original text.

[Fig 3.6: Recovered file]

OBSERVATION:

A test file was created and then deleted from the USB drive. FTK Imager was used to image the drive, and the hashes matched on verification. The recovery tool scanned the image for the deleted file. Whether recovery succeeded depended on the file system, the imaging method and whether the data was overwritten.

RESULT:

Thus, deleted files were recovered from a forensic image using forensic tools and the recovered data was verified.

Experiment 4 – Find Last Connected USB on Your System (USB Forensics)

AIM:

To find the USB devices used on a computer and study their details with USBDeview.

SOFTWARE REQUIREMENTS:

- OS: Windows
- Tool: USBDeview
- A system on which USB devices have been used before

THEORY:

Windows keeps a record of every USB device that has been plugged into it. USB forensics uses these records to show which devices were connected and at what time, which is useful when data theft through pen drives is suspected.

USBDeview reads these records and displays connected as well as previously connected devices. It shows fields such as:

- Device name and description
- Device type
- Serial number (when available)
- First and last connection times
- Vendor ID (VID) and Product ID (PID)

PROCEDURE:

1. Download and open USBDeview.

2. Wait for it to load all USB records.
3. Scroll through current and past devices (connected ones are highlighted).
4. Select the device to be examined.
5. Check its serial number, VID and PID.
6. Note the connection date and time fields.
7. Record the details.
8. Capture screenshots of the list and the device Properties.

[Fig 4.1: List of USB devices]

[Fig 4.2: Device Properties]

OBSERVATION:

USBDeview showed both present and past USB devices along with their type, VID, PID, serial number and timestamps. These details helped trace USB usage on the system.

RESULT:

Thus, USB connection history was examined using USBDeview and the device details were recorded for forensic analysis.

Experiment 5 – View Last Activity of Your PC

AIM:

To study recent user and system activity on a Windows PC using LastActivityView.

SOFTWARE REQUIREMENTS:

- OS: Windows
- Tool: LastActivityView

THEORY:

LastActivityView collects activity traces from different Windows sources such as Prefetch, the registry and event logs, and shows them as one time-ordered list.

The list can include programs executed, files and folders opened, system boot and shutdown, and network events. The amount of history shown depends on how many traces remain on the system.

PROCEDURE:

1. Download and run LastActivityView.
2. Allow it to read the system records.
3. Go through the list of activities.

4. Observe Action Time, Description, Filename, Full Path and Data Source for each entry.
5. Find the entries relevant to recent use.
6. Note the findings.
7. Capture screenshots of the list and one entry's Properties.

[Fig 5.1: Activity list]

[Fig 5.2: Entry Properties]

OBSERVATION:

The tool listed recent actions and system events with their times and sources, giving a clear picture of how the computer had been used recently.

RESULT:

Thus, recent computer activity and system events were examined using LastActivityView for digital forensic investigation.

Experiment 6 – Extraction of Browser Artifacts

AIM:

To collect and analyse Chrome browser artifacts – cache and history – using ChromeCacheView and ChromeHistoryView.

SOFTWARE REQUIREMENTS:

- OS: Windows
- Browser: Google Chrome
- Tools: ChromeCacheView, ChromeHistoryView

THEORY:

Every time a user browses, Chrome stores data such as cached page resources and a history of visited URLs. These are called browser artifacts and can show an investigator what a user did online.

ChromeCacheView opens Chrome's cache and shows each stored item with its URL, content type, size and access time.

ChromeHistoryView opens Chrome's history database and shows visited pages with their URL, title, visit time, visit count and referrer.

PROCEDURE:

A. Using ChromeCacheView

1. Download and run ChromeCacheView.

2. Let it load Chrome's cache.
3. Look through the cached entries.
4. Note URL, content type, file size and access time.
5. Mark the entries of interest.
6. Capture a screenshot.

[Fig 6.1: Cache entries]

B. Using ChromeHistoryView

1. Download and run ChromeHistoryView.
2. Let it load Chrome's history.
3. Look through the visited pages.
4. Note URL, title, visit time and visit count.
5. Open Properties of one record for referrer and transition details.
6. Capture screenshots.

[Fig 6.2: History records]

[Fig 6.3: Record Properties]

OBSERVATION:

S. No.	Tool	Artifact	Observation
1	ChromeCacheView	Cache	Cached item details were shown
2	ChromeHistoryView	History	Visited page records were shown

RESULT:

Thus, browser cache and history artifacts were extracted and analysed using ChromeCacheView and ChromeHistoryView.

Experiment 7 – Comparison of Two Files for Forensic Investigation Using Compare It Software

AIM:

To find the differences between two files using Compare It as part of a forensic investigation.

SOFTWARE REQUIREMENTS:

- OS: Windows

- Software: Compare It
- Two text files for comparison

THEORY:

Comparing files helps investigators detect tampering or track changes made to a document. Compare It loads two files next to each other and colours the lines that were inserted, removed or changed. This makes even small edits easy to spot.

PROCEDURE:

1. Install Compare It.
2. Create two text files with nearly identical content but a few altered lines.
3. Start Compare It.
4. Open the first file.
5. Open the second file.
6. View both files side by side.
7. Check the coloured lines for differences.
8. List the added, deleted and changed lines.
9. Save screenshots of the comparison.

[Fig 7.1: Side-by-side view]

[Fig 7.2: Differences report]

OBSERVATION:

The two files were shown together and every changed line was highlighted, clearly revealing the modifications.

RESULT:

Thus, two files were compared using Compare It and their differences were identified for forensic examination.

Experiment 8 – Collecting Email Evidence in Victim PC

AIM:

To acquire the RAM of a computer using DumpIt and analyse the memory image to find email-related evidence.

SOFTWARE REQUIREMENTS:

- OS: Windows
- Tool: DumpIt

- Analysis tool: bulk_extractor

THEORY:

Email forensics aims to find traces of email communication on a system. Volatile memory (RAM) holds live data of running programs, so webmail opened in a browser can leave addresses and message fragments in memory.

DumpIt is a one-click utility that writes the full physical memory of a Windows PC to a dump file. The dump is then scanned with **bulk_extractor**, which pulls out features such as email addresses, URLs and domains into text files. The dump should be hashed and stored carefully to protect its integrity.

PROCEDURE:

A. Memory capture with DumpIt

1. Copy DumpIt to the Windows system.
2. Run `DumpIt.exe` with administrator rights.
3. Check the destination path shown on screen.
4. Enter **y** to begin capturing.
5. Wait for the capture to end.
6. Find the `.dmp` file in the destination folder.
7. Note its name and size.

[Fig 8.1: DumpIt acquisition window]

B. Searching for email evidence

1. Open bulk_extractor and select the memory dump as input.
2. Choose an output directory and enable the email scanner.
3. Run the scan.
4. Open `email.txt` to view extracted addresses.
5. Open `email_histogram.txt` to see their counts.
6. Save the dump and outputs as evidence.

[Fig 8.2: bulk_extractor settings]

[Fig 8.3: Output files]

[Fig 8.4: email.txt / email_histogram.txt]

OBSERVATION:

The physical memory was successfully captured using DumpIt. Scanning the dump generated feature files, and the email output listed addresses found in RAM along with how often each appeared.

RESULT:

Thus, email evidence was collected from a computer by capturing its RAM with DumpIt and analysing the memory dump.

Experiment 9 – Hiding and Extracting a Text File Behind an Image/Audio File Using Command Prompt

AIM:

To conceal a text file inside an image file and an audio file using Command Prompt (with WinRAR for the audio method) and then extract the hidden content.

SOFTWARE REQUIREMENTS:

- OS: Windows
- Command Prompt
- WinRAR
- `B.jpeg`, `sound.mp3`, `A.txt`, `secret.txt`

THEORY:

The `copy /b` command can join two files byte by byte into a new file. Programs that open images or audio read only the data they need from the beginning of the file and skip anything added after it. So the new file still looks and plays like the original but secretly carries extra data.

For audio, the text file is zipped first. ZIP readers locate the archive from the end of the file, so WinRAR can open the combined MP3 and pull out the hidden file. Investigators detect this by checking unusual file sizes and the bytes after the media data ends.

PROCEDURE:

A. Text behind an image

1. Make a folder `hiding_text_in_image` on the Desktop.
2. Put `B.jpeg` and `A.txt` in it.
3. Open Command Prompt.
4. Go to the folder:

```
cd hiding_text_in_image
```

5. Combine the files:

```
copy /b B.jpeg + A.txt C.jpeg
```

Where:

- `B.jpeg` – source image
- `A.txt` – text to hide
- `C.jpeg` – output image with hidden text
- `/b` – binary mode

6. Check for the "1 file(s) copied" message.
7. Open `C.jpeg` to confirm it displays like the original.
8. Open `C.jpeg` with Notepad and check the end for the text.

[Fig 9.1: Folder contents]

[Fig 9.2: Command output]

[Fig 9.3: Hidden text visible in Notepad]

B. Text behind an audio file

1. Write some text in Notepad and save it as `secret.txt`.
2. Compress it to `secret.zip` using WinRAR.
3. Place `sound.mp3` and `secret.zip` in the `Steganography` folder.
4. Open Command Prompt in that folder.
5. Combine the files:

```
copy /b sound.mp3+secret.zip newfile.mp3
```

Where:

- `sound.mp3` – source audio
- `secret.zip` – archive with the text file
- `newfile.mp3` – output audio with hidden archive
- `/b` – binary mode

6. Confirm `newfile.mp3` is created and plays normally.
7. Optionally view it in Notepad to see the ZIP data at the end.
8. Open a copy of `newfile.mp3` in WinRAR, extract it, and verify `secret.txt`.

[Fig 9.4: Command output]

[Fig 9.5: File sizes of sound.mp3 and newfile.mp3]

[Fig 9.6: WinRAR extraction]

[Fig 9.7: Extracted files]

RESULT:

Thus, a text file was hidden inside an image using Command Prompt to create `C.jpeg`, and a zipped text file was hidden inside an audio file using WinRAR and Command Prompt to create `newfile.mp3`, from which the hidden file was recovered.

Experiment 10 – Extract EXIF Data from Image Files Using ExifReader Software

AIM:

To read and analyse the EXIF metadata stored in an image using ExifReader.

SOFTWARE REQUIREMENTS:

- OS: Windows
- Software: ExifReader
- A JPEG image from a mobile phone or camera

THEORY:

EXIF (Exchangeable Image File Format) is metadata embedded in photos by the capturing device. It commonly records:

- **Make and model** of the camera or phone
- **Date and time** of capture
- **Exposure settings** – shutter speed, F-number, ISO
- **Flash status**
- **Resolution** – width and height
- **GPS location**, when enabled

ExifReader opens an image and shows every EXIF field it contains. Investigators use this to connect an image with a device, time and possibly a location.

PROCEDURE:

1. Download ExifReader.
2. Open `ExifRead.exe`.
3. Click **Open**.
4. Go to the image location.
5. Select a JPEG/JPG file.
6. Confirm the thumbnail is shown.
7. Read the EXIF fields listed.
8. Note make, model, date/time, resolution, exposure, F-number, ISO and flash.
9. Record the useful metadata.

10. Capture a screenshot of the output.

[Fig 10.1: ExifReader output]

OBSERVATION:

ExifReader showed the image along with its EXIF details, such as device information, capture time and camera settings.

RESULT:

Thus, EXIF metadata of an image file was extracted and analysed using ExifReader.